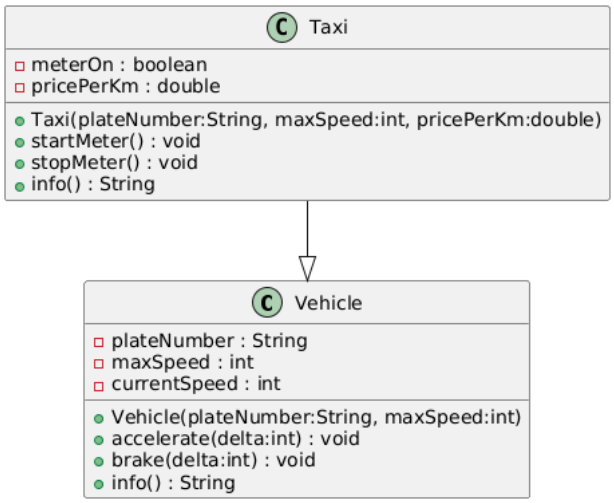
Lab 05

Task 1: Implement a base `Vehicle` class and a specialized `Taxi` class. Use `super(...)` to initialize shared vehicle attributes and `info()` to include taxi-specific state. Demonstrate speed control (accelerate/brake) and taxi meter toggling in `Main`.



Task 2: Filesystem

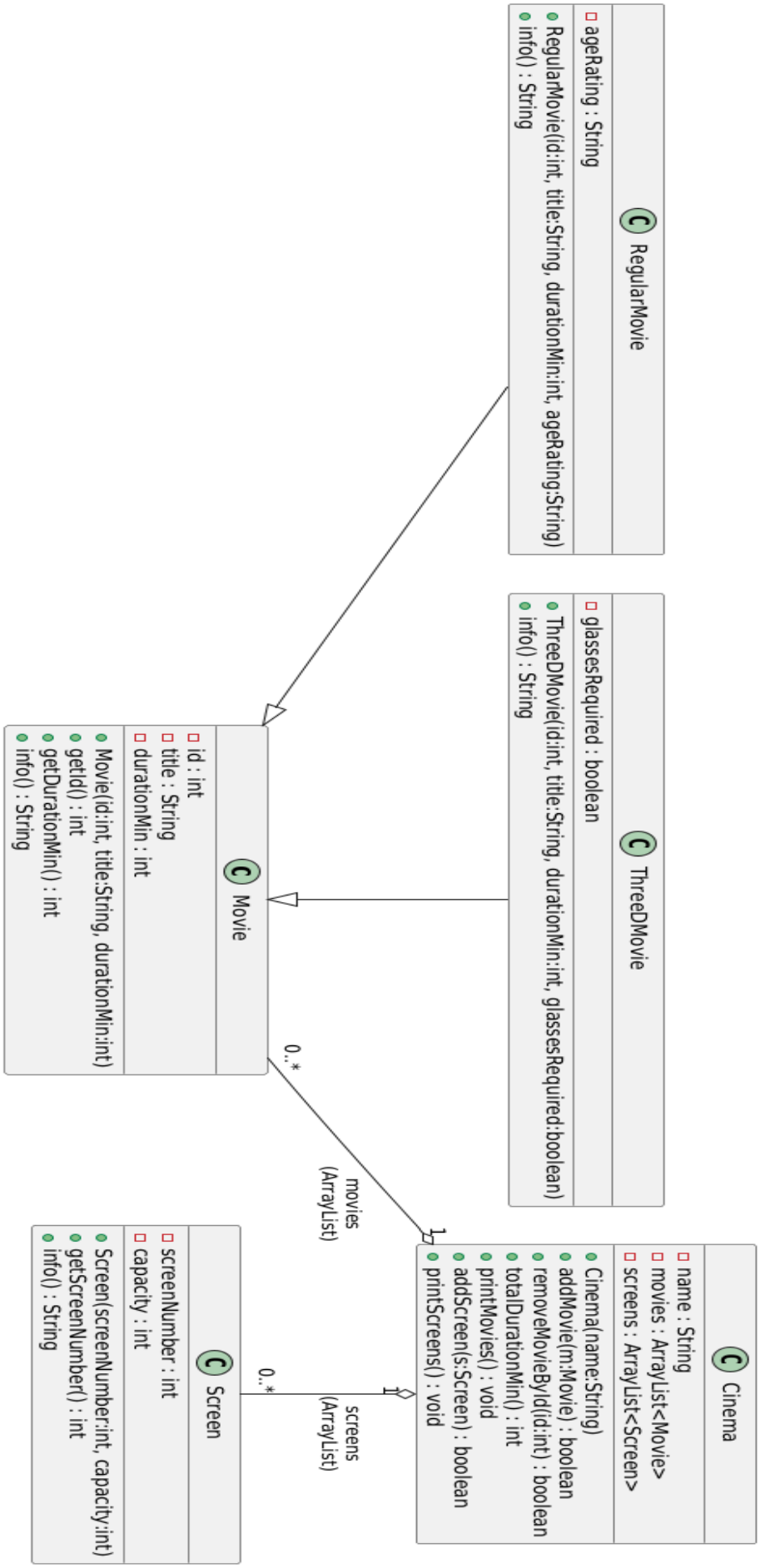
Create a filesystem abstraction with a parent `FsItem` and two subclasses `TextFile` and `ImageFile`. Each subclass must call `super(...)` to initialize common fields and `info()` to display its additional metadata.

- Provide a class diagram
- In `Main`, test renaming and validation with both valid and invalid inputs.

Task 3: Cinema Booking System

Build a cinema model based on the provided class diagram. In `Main`, create a cinema instance, add and remove movies/screens, and compute the total duration of all movies.

Dr. Oussama SERHANE (o.serhane@esi-sba.dz)


RegularMovie
ageRating : String
RegularMovie(id:int, title:String, durationMin:int, ageRating:String)
info() : String
ThreeDMovie
glassesRequired : boolean
ThreeDMovie(id:int, title:String, durationMin:int, glassesRequired:boolean)
info() : String
Movie
id : int
title : String
durationMin : int
Movie(id:int, title:String, durationMin:int)
getId() : int
getDurationMin() : int
info() : String
0..*
movies
(ArrayList)
1
Cinema
name : String
movies : ArrayList<Movie>
screens : ArrayList<Screen>
Cinema(name:String)
addMovie(m:Movie) : boolean
removeMovieById(id:int) : boolean
totalDurationMin() : int
printMovies() : void
addScreen(s:Screen) : boolean
printScreens() : void
1
screens
(ArrayList)
0..*
Screen
screenNumber : int
capacity : int
Screen(screenNumber:int, capacity:int)
getScreenNumber() : int
info() : String